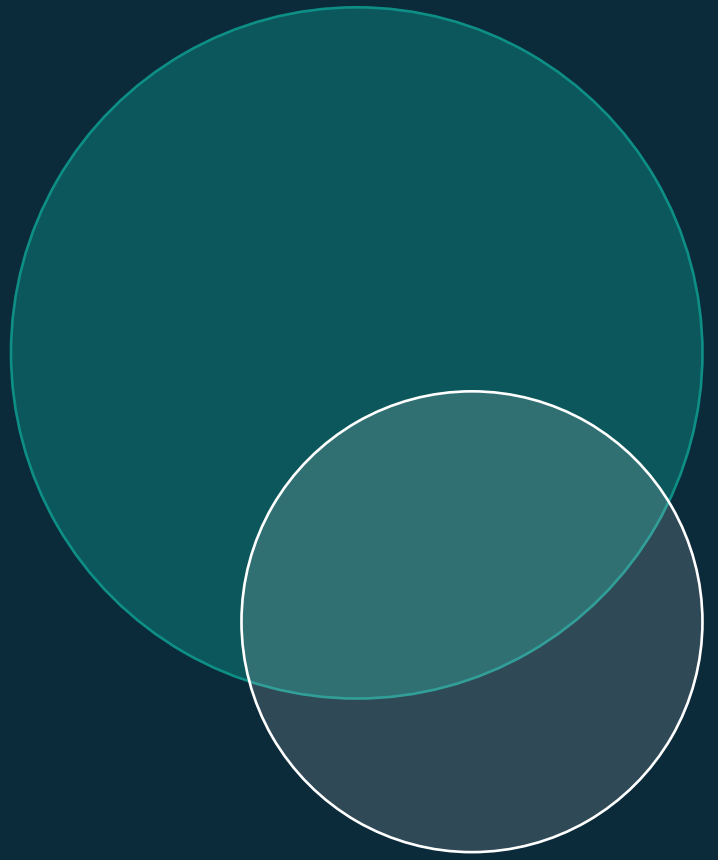
# GlucoTwin

## An explainable, personalized glucose Digital Twin for type 2 diabetes care

Digital Twin Challenge 2026 | research proof of concept

*[TEAM NAME / INSTITUTION / MEMBERS - to be filled before submission]*



All demonstration patients and all reported results are SYNTHETIC. Not clinically validated. Not a medical device.

# Healthcare problem: diabetes at national scale

Type 2 diabetes dominates; day-to-day glucose between clinic visits is largely invisible

**~101 million**

people with diabetes in India (11.4% of the population), ICMR-INDIAB estimate

**~136 million**

people with prediabetes (15.3%), a pool that can convert to diabetes

**113,043**

survey participants across states and UTs, 2008-2020

**Why forecasting + a twin?**

Continuous glucose monitors produce a reading every few minutes, but clinicians see snapshots. A continuously updated, explainable patient state with honest uncertainty could help a clinician review where a patient is heading. This is our motivation, not a demonstrated benefit.

*Source: Anjana et al., Lancet Diabetes Endocrinol 2023 (DOI 10.1016/S2213-8587(23)00119-5), figures as reported by secondary coverage and a Natl Med J India commentary; verify against the paper.*

# Existing solutions and literature

What we found - and what we could not verify

### Digital twins for health

Scoping reviews (e.g. Katsoulakis et al., npj Digit Med 2024) survey health digital twins and their applications. We use the working idea of a personalised, continuously updated, predictive patient model.

### CGM forecasting

Published work forecasts glucose from CGM history (e.g. GARNN, arXiv 2402.16230; a Bayesian T2DM approach, arXiv 2409.07315). We did not reproduce them; we benchmark only against persistence and ridge.

### Open datasets

DiaTrend: type 1, controlled access. ShanghaiT2DM: type 2, 15-min CGM, public but licence terms conflict across sources. Glucose-ML (2025) surveys these collections.

### Other challenge entries

Public repositories exist for this challenge, including one already named 'GlucoTwin' and one 'ThaliTwin'. We did not audit their claims; we flag the name conflict.

# Research gaps and our contribution

A small, carefully evaluated proof of concept

### Gaps we target

Leakage-resistant, patient-level evaluation against simple baselines

Explicit behaviour for stale, missing and corrupt input (abstain, not guess)

Explanations that are faithful to the model, with causal limits stated

What-if tools that do not invent physiology

### What this repository contributes

Twin state engine: validation, quarantine, forecast history, audit, restart-safe

Per-horizon forecasts (15-120 min) with conformal intervals

Exact additive explanations and revision tracking

Scenario lab limited to supported inputs

54 tests; reproducible results with ablations reported even when negative

# Problem statement and intended outcome

One localised outcome, not the whole body

### Primary task

Given the last 60 minutes of CGM readings (15-min grid) plus patient history, forecast glucose for the next 15-120 minutes with an uncertainty interval, an explanation and a data-quality status.

### Intended outcome

A working data-to-forecast loop that a clinician-facing dashboard can show, test and audit - and an honest benchmark.

### Explicitly out of scope

Diagnosis, treatment or insulin/medication advice. Alarms. Any claim of clinical validation.

# Data sources, privacy and limitations

Research found hard limits; we documented a safe fallback

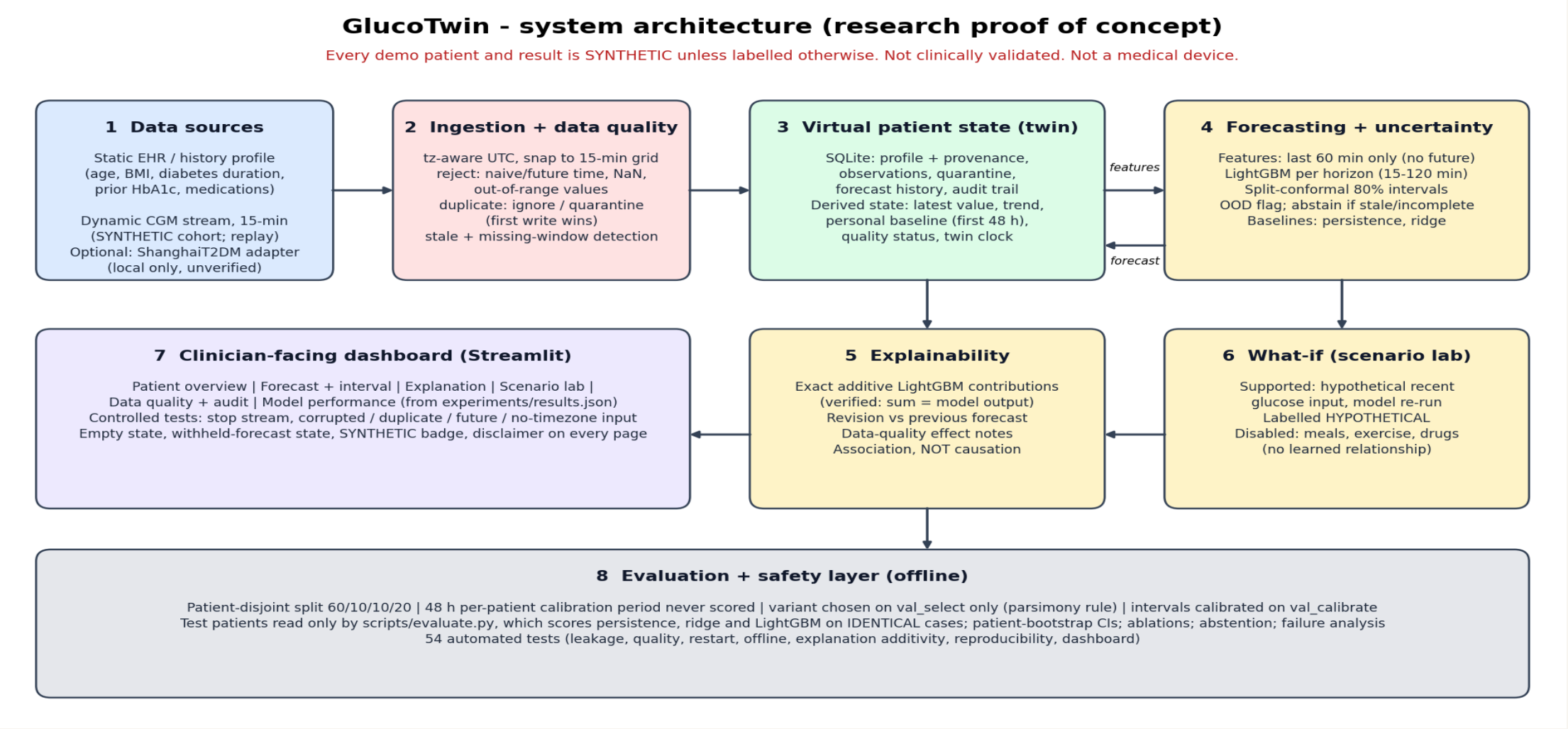
| Dataset | Population | Access finding | Used? |
|---|---|---|---|
| DiaTrend | Type 1, CGM | Controlled access (Synapse + agreement); no redistribution | No |
| ShanghaiT2DM | Type 2, 15-min CGM | Public, but licence terms conflict across sources; host unreachable from the build sandbox | Adapter only (untested on real files) |
| Synthetic cohort | T2D-like, 100 pts x 14 d | Generated here, seeded, MIT, labelled SYNTHETIC on every record | Yes - all results |

- No real patient data in the repository, logs or figures
- Static-to-glucose links in the synthetic data exist by construction: they cannot validate physiology
- Real-data results must be produced and reported separately

**No result in this deck says anything about real patients.**

# Architecture and data fusion

History + stream -> quality gate -> twin state -> forecast -> explanation -> dashboard



# Dynamic update workflow and dashboard

Every new reading updates state, forecast, explanation and audit trail

1 **New reading**

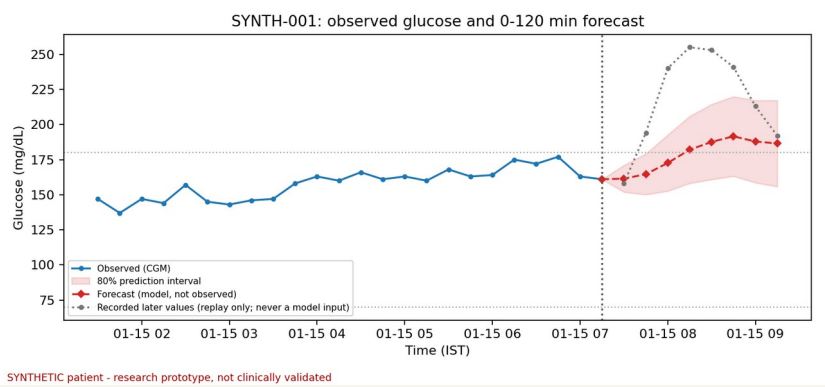
2 **Validate**

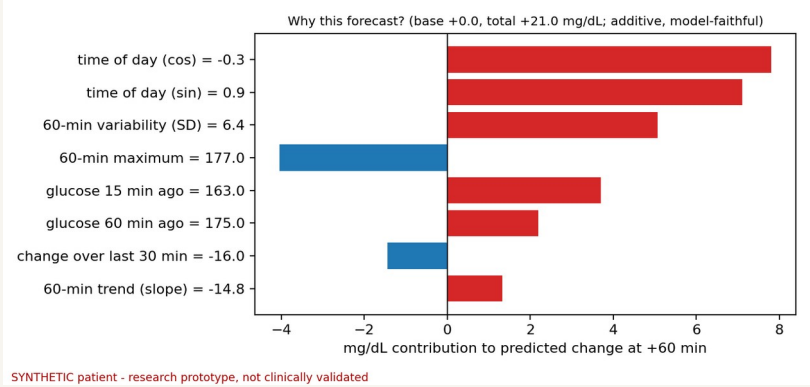
3 **Append / quarantine**

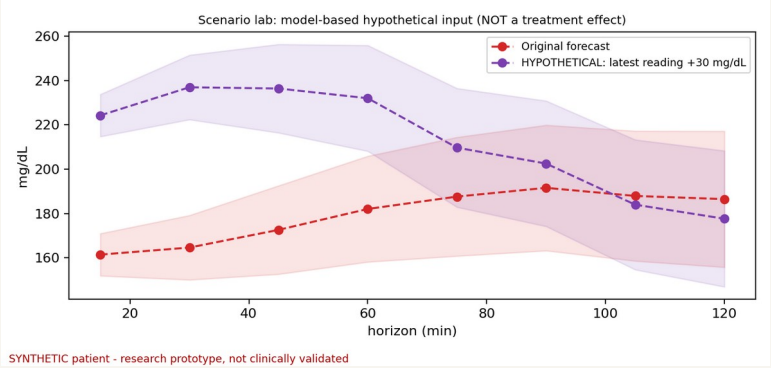
4 **Quality check**

5 **Forecast or withhold**

6 **Explain + audit**



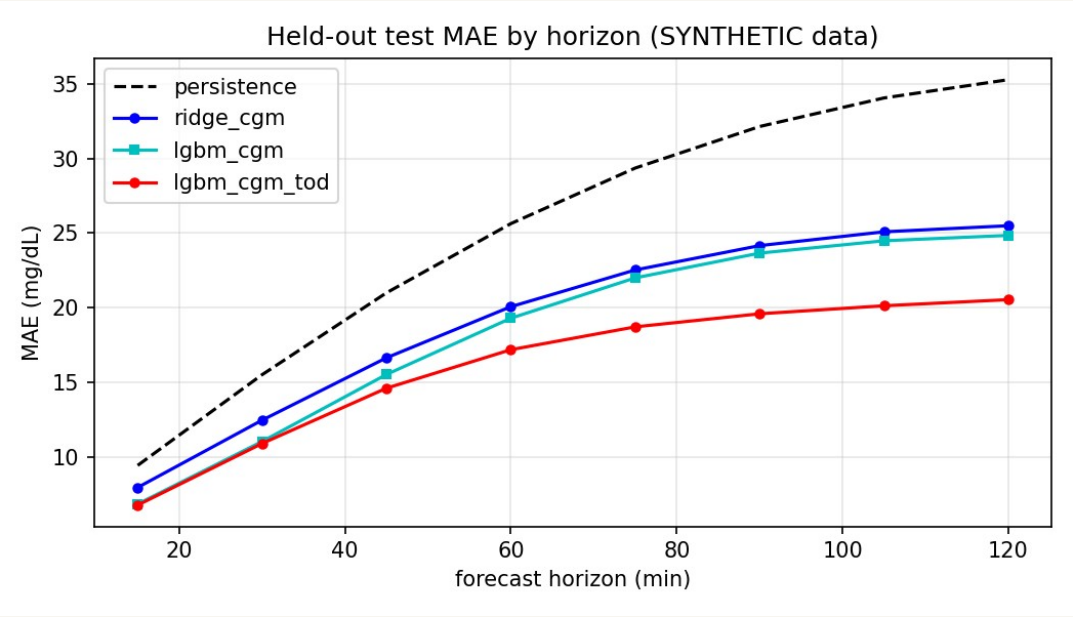




- Six tabs: overview, forecast, explanation, scenario lab, quality + audit, model performance
- Explanations are exact additive terms of the model output; scenario lab is labelled HYPOTHETICAL, meal/exercise/drug controls disabled
- Replay of a recorded SYNTHETIC stream; there is no live-device connection

# Model evaluation and baseline comparison

Held-out SYNTHETIC test: 20 patients, 20,825 windows, identical cases for all models




| MAE mg/dL | +15 min | +30 min | +60 min | +120 min |
|---|---|---|---|---|
| Persistence | 9.5 | 15.5 | 25.6 | 35.3 |
| Ridge | 8.0 | 12.5 | 20.1 | 25.5 |
| LightGBM (CGM only) | 6.9 | 11.1 | 19.3 | 24.8 |
| **LightGBM + time of day** | **6.8** | **10.9** | **17.2** | **20.5** |

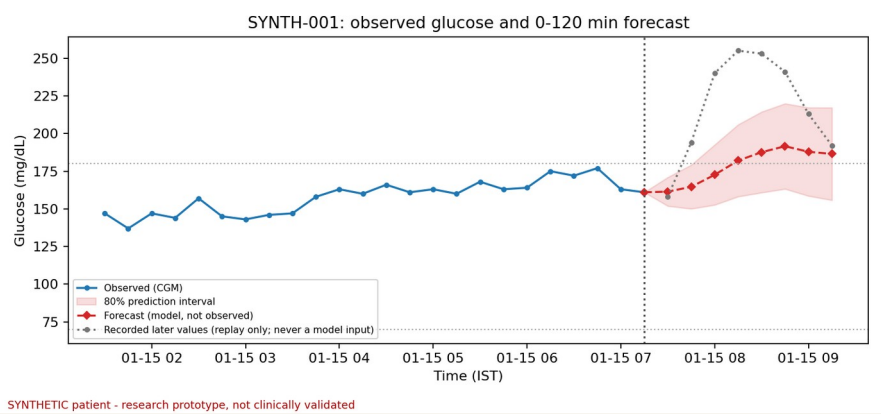
- Primary vs persistence at +60 min: -8.44 [-11.12, -6.24] mg/dL (95% patient-bootstrap CI); better for 20 of 20 patients
- Ablations vs CGM-only at +60 min: time of day -2.11 [-2.70, -1.47] (expected by construction: synthetic meals are scheduled); personal baseline -0.05 [-1.02, 1.04]; static EHR +0.05 [-0.73, 0.86] - no reliable gain at +60 min (personal baseline helps only ~0.3 mg/dL at +15 min)

**Synthetic data: shows the pipeline works and recovers the generator's dynamics - not clinical accuracy. Preliminary.**

# Tests, data-quality behaviour and uncertainty

Fail safe, show uncertainty, admit misses




- Stale (> 20 min) or incomplete 60-min window: forecast withheld; 6.4% of observed test origins (synthetic missingness)
- Rejected / quarantined: naive or future timestamps, NaN, out-of-range values, conflicting duplicates
- 80% intervals: coverage 78-80% on test; per patient 71-85% (marginal guarantee only)
- Hardest case: flat glucose before a meal (MAE 22.6 vs 12.1 mg/dL rising, +120 min); bottom-left: a breakfast spike escaping the interval
- 54 automated tests incl. leakage, restart, offline, additivity, dashboard; mutation-checked

# Limitations, responsible use and next steps

**Not claimed**

Clinical validation or real-patient accuracy

Benefit for type 2 populations (data are synthetic)

Causal effects of food, exercise or drugs

Use for diagnosis, treatment or medication decisions

That time-of-day or EHR gains transfer to real data

**Next steps**

1. Run the ShanghaiT2DM adapter locally; verify columns, units, licence
2. Re-train and evaluate; report real-data results separately
3. Add meal/activity inputs only if real data support them
4. Clinician review and prospective evaluation before any clinical use
5. Resolve the project-name conflict

# Conclusion

- A reproducible twin pipeline: fused history + stream, versioned forecasts, explanations, safe withholding
- On held-out synthetic patients it beats persistence and ridge; value on real patients is untested
- Negative and null findings are reported alongside positives

*Repository: [PUBLIC GITHUB URL - to be filled]*
*Demo video (20+ min, unlisted): [YOUTUBE URL - to be filled]*
*Team / college / incubator: [to be filled]*

Research proof of concept. Synthetic data. Not clinically validated. Not a medical device.